

# BLACKBOX

## NETWORK INCIDENT INVESTIGATOR

Five PCAP tests · Simple workflow · Honest evidence

GitHub-ready release 1.5 | October 2026

Turn captured traffic into a readable investigation: inspect network communications, evaluate rule-based findings, trace packet references, and export case notes.

Three bundled lab captures were tested. Two additional public, documented PCAP sources are linked for independent testing and are not included or represented as already tested.

Accuracy note: A valid PCAP contains packet records, but the source or attack ground truth of the provided classroom files is not independently verified. A rule finding is a lead, not proof of compromise.

# 01 / Use BLACKBOX in five steps

A simple path through the app. The advanced tools remain available without cluttering the main navigation.

### 01 Open an investigation

Select Test real captures to load the two bundled lab cases, or create your own case.

### 02 Import evidence

For a personal case, open Evidence vault, import an authorized .pcap or .pcapng, and check the SHA-256 and import status.

### 03 Review findings

Inspect the Overview and Findings. Click a finding to see the linked packet IDs and what triggered the rule.

### 04 Verify in context

Use Network map to check who spoke with whom. Open Packet explorer or Incident replay under More tools for depth.

### 05 Explain and export

Ask the Investigator an offline evidence question and export the Case report. Cloud AI is optional and not used publicly.

## What the labels mean

Ready = packet parsing completed within supported limits. Partial = not all evidence was decoded. Finding = configured heuristic condition crossed its threshold. Zero findings never means “safe.”

Built-in evidence tools are deterministic. A separate provider key is required for external model-generated explanations; the trained ML model from the teammate is not yet integrated.

# 02 / Two ready-to-open lab investigations

Use these for the 2–4 minute Thirduni screen recording. The public deployment contains two preloaded, read-only cases; local mode allows your own uploads.

## LAB A / ARP consistency

278 frames parsed | 2 heuristic findings

Two medium-severity ARP identity-rule findings for 192.168.56.10 and 192.168.56.20.

Show on screen: Open Findings, inspect the packet references, then explain why conflicting ARP claims are not conclusive evidence of an attack.

## LAB B / DNS and DHCP

304 frames parsed | 0 heuristic findings

TCP, UDP, ICMP and ARP packets are visible; no configured rule threshold was crossed.

Show on screen: Switch cases, inspect the protocol mix, ask the offline Investigator what is observable, and explain that zero findings is not a clean bill of health.

## Suggested 25-second voiceover

“BLACKBOX reads network captures and explains what we can actually observe. Here is one case with ARP inconsistencies, and a second with normal DNS/DHCP traffic. Every finding links back to packets, and we can export the evidence for review.”

Never say: “We proved this computer was hacked” or “our ML model detected this” unless you have independently verified the attack and integrated that model.

# 03 / Five PCAP test resources

Three tested lab files are in this release. Two independently documented public captures are linked for you to download and test; no results are invented for those sources.

## 01. Legacy service traffic

TESTED · 278 frames / 1 findings

TCP/21 signal and normal protocol traffic; rule alert does not prove exploitation.

File: `samples/recorded_labs/lab_legacy_service.pcap`
SHA-256: `9bdf0792da1c3048f21e8cce4387b52a599e053dc3feb157fb9a44bf2b0d3eb7`

## 02. ARP identity inconsistencies

TESTED · 278 frames / 2 findings

Investigate ARP MAC/IP disagreements; this is a heuristic, not proof of spoofing.

File: `samples/recorded_labs/lab_arp_observations.pcap`
SHA-256: `160bc3e92c4311dcb9f06fa781d14d18d8ae83c3f2060944823cba46bbb04ced`

## 03. DNS and DHCP baseline

TESTED · 304 frames / 0 findings

Ordinary DNS/DHCP exchanges, no rules crossed in previous test.

File: `samples/recorded_labs/lab_dns_dhcp.pcap`
SHA-256: `a0a0c4fca426217d3066c1df5cbe37b18854d5b77ef5ed49586bdd00871e00a9`

## 04. Nmap SYN scan (independent public lab)

PUBLIC SOURCE · NOT TESTED HERE

Actual Nmap -sS lab scan; suitable for scanning indicators. Test in the app after downloading.

Open original download source

## 05. TCP/HTTP file transfer with ECN

PUBLIC SOURCE · NOT TESTED HERE

Documented HTTP file-transfer capture with explicit congestion notification. Test after downloading.

Open original download source

# 04 / Repeat the tests and publish safely

## Testing each capture

1. Run BLACKBOX on Windows. Create a separate case for each PCAP.
2. Import the file under Evidence vault. Record its SHA-256, parsing status and frame count.
3. View Overview and Findings; review evidence frames in Packet explorer and Wireshark.
4. Export the JSON or Markdown case report and compare the results.

## Source references

Public Nmap SYN lab: github.com/Ryan-Sapone/Nmap-and-Wireshark-Lab

Wireshark HTTP sample: wiki.wireshark.org/SampleCaptures (find tcp-ecn-sample.pcap)

The three local lab captures were supplied in the user's earlier network exercises; capture origins and redistribution permission were not independently verified.

## Validated release behavior

38 Python automated tests passed. All three bundled captures imported with status Ready. The read-only seeded gallery displayed two cases through the API, blocked write requests and returned deterministic offline explanations. Desktop browser screenshots could not be end-to-end verified in this environment.

## Publishing to Thirduni

Push source code to GitHub only after confirming the lab PCAPs are safe to redistribute. Deploy a separate public-demo site using the Dockerfile, set `BLACKBOX_DEPLOY_MODE=public-demo` and your actual `BLACKBOX_PUBLIC_HOST`, and keep the gallery read-only. Give judges a public HTTPS address, test every link while logged out, then record a 2–4 minute video showing the working cases.

Limitations: Rule-based alerts are not ML predictions. No real-time monitoring, decryption, or verified attacker attribution. Treat network metadata as potentially sensitive.